Collective Critics for Creative Story Generation

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Abstract

Generating a long story of several thousand words with narrative coherence using Large Language Models (LLMs) has been a challenging task. Previous research has addressed this challenge by proposing different frameworks that create a story plan and generate a long story based on that plan. However, these frameworks have been mainly focusing on maintaining narrative coherence in stories, often overlooking creativity in story planning and the expressiveness of the stories generated from those plans, which are desirable properties to captivate readers' interest. In this paper, we propose Collective **Criti**cs for **C**reative **S**tory Generation framework (CRITICS), which is composed of plan refining stage (CRPLAN) and story generation stage (CRTEXT), to integrate a collective revision mechanism that promotes those properties into long-form story generation process. Specifically, in each stage, a group of LLM critics and one leader collaborate to incrementally refine drafts of plan and story throughout multiple rounds. Extensive human evaluation shows that the CRITICS can significantly enhance story creativity and reader engagement, while also maintaining narrative coherence. Furthermore, the design of the framework allows active participation from human writers in any role within the critique process, enabling interactive human-machine collaboration in story writing.[1]

1 Introduction

Recent progress in Large Language Models (LLMs) has enabled the automatic generation of long-form stories containing several thousand words. This advancement has led to the proposal of various frameworks (Yang et al., 2022, 2023; Zhou et al., 2023) that are capable of generating stories maintaining narrative coherence. In pursuit of this capability, many frameworks adopt a two-step approach (Fan et al., 2018; Yao et al., 2019; Fang et al., 2021; Tian and Peng, 2022): (a) creating a story plan; and (b) generating a detailed story based on the plan. To explore further, some works have made effort to improve structure of story plan (Wang et al., 2023b) and refine the writing style (Kong et al., 2021; Zhu et al., 2023b). Despite their ability to produce consistent and coherent text, these works frequently fall short in crafting creative long-range narratives that captivate human interest, such as stories with emotional flow, flashback structures, and other engaging elements (Zhang et al., 2021; Huang et al., 2023b).

Therefore, we propose Collective **Criti**cs for **C**reative **S**tory Generation (CRITICS), a novel framework for long-form story generation that integrates collaborative critique into the plan-to-story framework. Our framework focuses on creatively refining story plans (CRPLAN) and enhancing the expressiveness of stories (CRTEXT). Inspired by studies demonstrating that collaborative work significantly enhances creativity through the sharing and integration of knowledge and ideas from various fields (Mamykina et al., 2002; Barrett et al., 2021), CRITICS adopts a collaborative critique approach. Specifically, in each stage, multiple LLM critics evaluate the draft and provide suggestions for improvement based on criteria (e.g., uniqueness of the narrative flow and vividness of the generated text) which is designed for assessing creativity (Mozaffari, 2013). Then, a leader selects a critique that best helps refine the draft. This revision process iterates through multiple rounds to produce a final plan and story (Figure 1).

In CRPLAN, the critics focus on infusing originality into the story plan (e.g., unexpected shifts in the story's atmosphere, twisted endings). To do so, critics are endowed with personas that are adaptive to a given storyline, which helps make clear and diverse suggestions to enhance the story's creativity

[1]Our code are publicly available at https://github.com/EMNLP-2024-CritiCS/Collective-Critics-for-Creative-Story-Generation

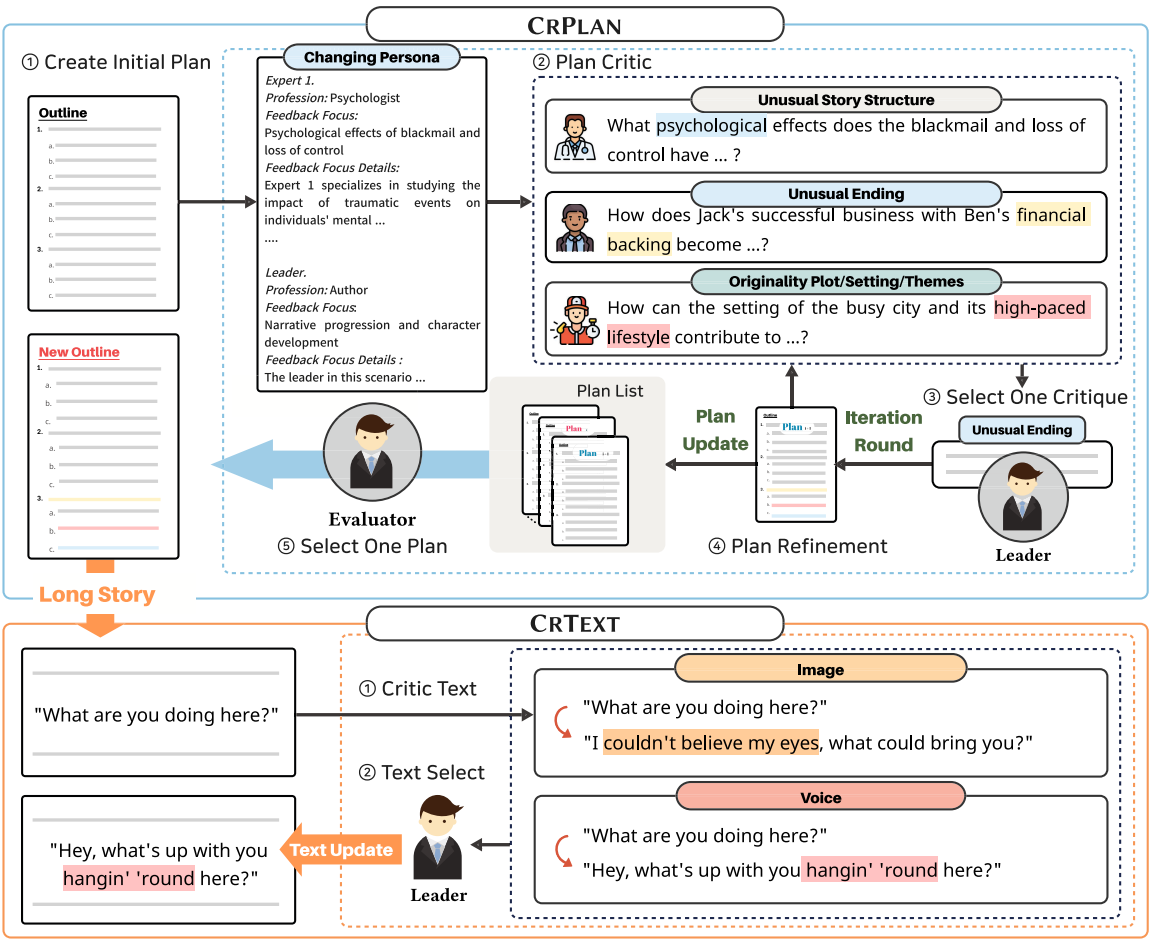

Figure 1: The framework comprises two stages: CRPLAN and CRTEXT. CRPLAN involves five phases: creating a story plan from a premise, reviewing the plan with critics' persona-driven perspectives, selecting a critique for revision by a leader, storing the revised plan, and choosing a plan for further development by an evaluator. Personas of critics are created based on the themes or content of the narratives, which helps in generating detailed and contextually relevant critiques. Please refer to Appendix A.1 critiques and refines long story text based on creative criteria, with a leader selecting and improving the best expressions.

and coherence. CRTEXT is designed to improve the expressiveness of stories. It refines the narrative to incorporate unique expressions and vivid descriptions. This includes comic book onomatopoeias (e.g., '*swooosh*'), figurative phrases (e.g., '*silent twilight*'), and detailed sensory experiences (e.g., a '*buzzing market*' or the '*sharp tang of sea air*').

Extensive human evaluation demonstrates that our CRITICS produces more creative and thus interesting plan, and generates expressive story. To be specific, CRITICS outperforms the state-of-the-art method on the metrics of creativity and interestingness by a large margin. Additionally, detailed analyses indicate that our design choices for CRITICS (e.g., applying multiple criteria, role of leader critics, and bestowing personas)

In addition, CRITICS facilitates interactive human-machine collaboration in writing by enabling humans to participate as any of the players within the framework, refining stories according to their preferences.

The main contributions of this study are:

1. We propose a framework named CRITICS, which is designed to enhance story plans and text expressiveness creatively through collective critiques based on criteria of creativity.

2. Extensive human evaluation and ablation studies demonstrate that CRITICS effectively creates stories with creativity and coherence, employing key components such as diverse criteria, leaders, and personas.

3. CRITICS supports interactive writing, enabling human users to intervene in critiques and actively refine stories.

2 Related Work

Long-form Story Generation. In prior research, the focus of automatic story generation is to create short stories composed of several completed sentences (Roemmele and Gordon, 2018; Ammanabrolu et al., 2020; Wang et al., 2022b). With the development of LLMs, studies on story generation have shifted their focus toward generating longer narratives of several thousand words, maintaining narrative coherence (Yang et al., 2022, 2023; Zhou et al., 2023). Those works have tried to sustain coherence in longer stories by employing story planning techniques such as hierarchical generation (Fan et al., 2018, 2019). This approach involves first crafting a broad story outline at a high level, which serves as the foundation for the detailed narrative development. Such hierarchical story plans typically consist of outlines with around 1K tokens, allowing for the creation of longer stories without losing coherence. Recent research focuses on creating robust story plans (e.g., story plans with high narrative coherence, story plans with appropriately placed event scenes) (Ammanabrolu et al., 2020; Zhu et al., 2023a; Wang et al., 2023b) and transferring plain text into literary expressions (Kong et al., 2021; Zhu et al., 2023b; Huang et al., 2023b). However, those works fall short in devising unusual, original stories that engage the reader's interest. To address these issues, in this study, we focus on the creativity of stories.

Creative Writing. The assessment of creativity in creative writing has been persistently discussed among researchers. Traditional approaches to assessing creativity have relied on criteria like novelty and unexpectedness (Barron, 1955; Rhodes, 1961; Blomer, 2011). However, these approaches are abstract, as they typically rely on broad standards for evaluation. The need for specific criteria to assess the writing has led to rubric-based methods using the traits of creative writing (Arter and McTighe, 2001; Young, 2009; Mozaffari, 2013), generally assessing creativity in three aspects (Mozaffari, 2013): *Image*, *Voice*, and *Originality*. In this study, we adopt those criteria to give more detailed guideline for critcs to refine stories.

Collaborative Writing. Research on collaborative writing as a method to produce creative authoring is actively ongoing (Csikszentmihalyi, 1997; Mamykina et al., 2002; Paulus et al., 2012). Compared to individual writing, collaborative writing can lead to more divergent works by sharing different knowledge (Bayle et al., 1998; Resnick et al., 2005; Sie et al., 2009; Bitter-Rijpkema et al., 2011). The more diverse the backgrounds, skills, and experience levels of the collaborators, the more varied the knowledge shared, enhancing the creativity of the writing. Inspired by this, we design CRITICS to accommodate multiple critics, each with unique characteristics, to increase diversity.

Interactive Agents. Recently, there has been a strong focus on developing interactive agents powered by LLMs (OpenAI, 2023; Team et al., 2023; Anil et al., 2023; OpenAI et al., 2024). These agents aim to enhance collaboration, enabling them to share information and solve complex problems together. Earlier studies (Yao et al., 2023; Du et al., 2023; Gou et al., 2024; Sun et al., 2024; Chen et al., 2024a) focused on improving the reasoning capabilities of LLM agents through interactions. However, recent findings (Huang et al., 2023a; Wang et al., 2024) suggest that their reasoning abilities cannot improve solely through interaction without external feedback. On the other hand, current research has shifted towards tasks with various possible solutions, such as code generation (Chen et al., 2023; Hong et al., 2023; Holt et al., 2023) and embodied AIs (Song et al., 2023; Wang et al., 2023a; Shek et al., 2024; Chen et al., 2024b), where agents cooperate to achieve specific goals. Our research utilizes interactive agents to build divergent critique processes for creative long-story generation. Integrating human and agent participants enriches narratives with diverse perspectives, making long stories more engaging and creative.

3 Collective Critics for Creative Story Generation

CRITICS is composed of two stages (CRPLAN and CRTEXT) that utilize the capabilities of LLM to write creative stories. Our framework enhances creativity at both the micro level (sentence expressiveness) and the macro level (story structure, themes, endings, etc.) To achieve this, we incorporate the critique process based on creative criteria (Mozaffari, 2013) into our framework. Although we commonly use criteria for assessing the creativity of stories in literature, our framework is designed to accommodate a variety of other criteria as well.